



Applications of machine learning methods for boiling modeling and prediction: A comprehensive review

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ABSTRACT

Boiling refers to the heat transfer mechanism that occurs due to the phase transition from liquid to vapor. In comparison with single phase heat transfer, this mechanism has several advantages such as much higher rate at lower temperature differences. Regarding the complexities of two phase heat transfer simulation, due to the involvement of various parameters in this phenomenon, applying intelligent methods such as artificial neural networks could be useful for modeling this type of heat transfer mechanism. In the present article, studies on the modeling of pool boiling heat transfer utilizing machine learning methods have been reviewed, and their findings are reflected. According to the outcomes of the reviewed works, it can be concluded that using intelligent methods can provide accurate predictions of pool boiling heat transfer with a R^2 of around 0.99 in some cases. In addition, by applying these methods it would be possible to predict the heat transfer in cases of utilizing nanofluids or porous media. The exactness and applicability range of these models is influenced by several elements such as the considered inputs, applied methods and employed functions. Using an appropriate method with optimal parameter values for the relevant intelligent method would lead to higher precision in modeling.

1. Introduction

In comparison with conventional heat transfer mechanisms without phase change, applying two phase heat transfer provides much higher rates of thermal energy transfer [1,2]. Through boiling it is possible to reach large heat transfer rates with small temperature differences. Two-phase heat transfer is of great interest in several applications including thermal management of electronic components, compact heat exchangers, processes in oil industry, biotechnology etc. [3–5]. Properties of the fluids used for two phase heat transfer, especially its latent heat, noticeably influences the heat transfer characteristics. Consequently, the performance of thermal devices working on the basis of two-phase heat transfer is greatly dependent on the specifications of the fluid. For in-

stance, in cases using heat pipes, as one of the efficient two-phase heat transfer mediums, fluids with very low boiling temperatures such as nitrogen, neon and hydrogen must be applied for very low operating states, cryogenic conditions [6], while conventional fluids such as water, ethanol and ethylene glycol are used in cases of medium temperatures [7,8] and fluids with higher boiling temperatures such as sodium are employed for high temperature thermal management, i.g. turbine blade cooling [9]. Latent heat of the fluid used in these devices would affect heat transfer capacity. Generally, higher values of latent heat would provide higher heat transfer capacity at the same conditions.

Besides the latent heat, two other factors including surface tension at the interface of liquid-vapor and the difference between the densities of two phases play key role in the rate of boiling heat transfer [10]. Nu-

Abbreviations: ACO, ant colony optimization; ANFIS, adaptive neuro-fuzzy inference system; ANN, artificial neural network; BP, back propagation; CF, cascade forward; CH-PSO, conjugate hybrid particle swarm optimization; DE, differential evolution; ELM, extreme learning machine; GA, genetic algorithm; GMDH, group method of data handling; GRNN, general regression neural network; HTC, heat transfer coefficient; LSSVM, least squared support vector machine; MAE, mean absolute error; MAPE, mean average percentage error; MARS, multivariate adaptive regression spline; MLP, multilayer perceptron; MLR, multiple linear regression; PSO, particle swarm optimization; RBF, radial basis function; RBN, radial basis network; RMSE, root mean squared error; SVM, support vector machine.

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cleus formation rate is proportional to $e^{-\sigma^3}$ [11], meaning higher rates of formation in cases of lower surface tensions. According to the correlations proposed for boiling heat transfer, increment in the density difference can result in higher heat flux [10]. Furthermore, some other factors such as the boiling surface influence the boiling heat transfer. For instance, Kim et al. [12] investigated the pool boiling heat transfer rate at the surface with moderate wettability. They noticed that the critical heat flux in case of the roughest surface was higher than the smoothest surface. In another study, Ahmad et al. [13] investigated the impact of surface roughness on the heat transfer rate of R-123 pool boiling. A significant enhancement in the heat transfer rate was observed in case of using a rough surface compared with a smooth surface. In addition, they applied an electrostatic field with high intensity and observed that the heat transfer rate increased under this condition. Sathyabhama and Prashanth [14] investigated impact of boiling surface vibration on the heat transfer. They found that induction of vibration can cause improvement in heat transfer coefficient by up to 26%, attributed to changes in thermal boundary layer due to the vibration.

Intelligent techniques are usable for forecasting and estimating the properties of materials and the behavior of complex systems [15–17]. In recent years, intelligent methods have been broadly applied in different fields of thermal engineering such as performance predictions of heat pipes and heat exchangers, modeling the properties of heat transfer fluids such as dynamic viscosity [18,19], specific heat [20] and thermal conductivity [21–23], and estimating the output of thermal systems such as solar collectors [24,25]. For instance, Ramezanizadeh and Alhuyi Nazari [26] used Artificial Neural Networks (ANN) to estimate the thermal conductivity of nanofluids composed of water and silver particles with different sizes and concentrations and found it to be extremely useful for obtaining accurate predictions. In another work [27], an ANN was applied to predict heat transfer in a circular tube with pins. It was found that the heat transfer, in terms of the Nusselt number, could be predicted with a high level of accuracy. Zolghadri et al. [28] employed ANN in order to forecast the energy consumption and Nusselt number of the heat exchanger with nanofluid in the processing of tomato paste and found that the proposed network in the optimal condition was able to estimate the mentioned parameters with high accuracy. Maddah et al. [29] applied ANN to predict the exergy efficiency of a heat exchanger and found the accuracy of their method significant with R^2 of 0.9967. In another work [30], different methods were used for predicting the output temperature of a nanofluid and Nusselt number of nanofluidic flow in a car radiator and observed that utilizing the MARS-linear method was more accurate in case of Nusselt number while an ANN outperformed other approaches for forecasting the output temperature.

Since the boiling is a multiscale phenomenon composed of various microscopic and macroscopic processes [31], its modeling is complicated and requires complex approaches. In addition, several factors affect these processes which cause difficulties in their simulation by using computational methods. Moreover, influences of some components involved in boiling process and its heat transfer specifications are not clear and linear that cannot be easily considered by the empirical correlations or the conventional approaches applied for estimation. In this regard, it is crucial to apply powerful approaches and methods for their accurate modeling. For this purpose, different intelligent methods such as ANNs are suggested. Concerning the performance dependency of these methods on different factors such as the algorithm, used function and architecture, it is crucial to get deep insight into these methods to propose proper models. In addition to the quantitative analysis of boiling phenomenon, some of its characteristics such as critical heat flux, transition of boiling regime etc. can be detected and analyzed by powerful tools proposed in recent years such as deep learning strategies or other machine learning methods [4,32,33]. In these types of studies, deep learning strategies could be used for analysis of the images obtained in the experimental tests related to the boiling [34]. For instance, Liu et al. [35] applied deep NN and simulation result to model the characteristics of boiling heat transfer. In their work, energy convective transport,

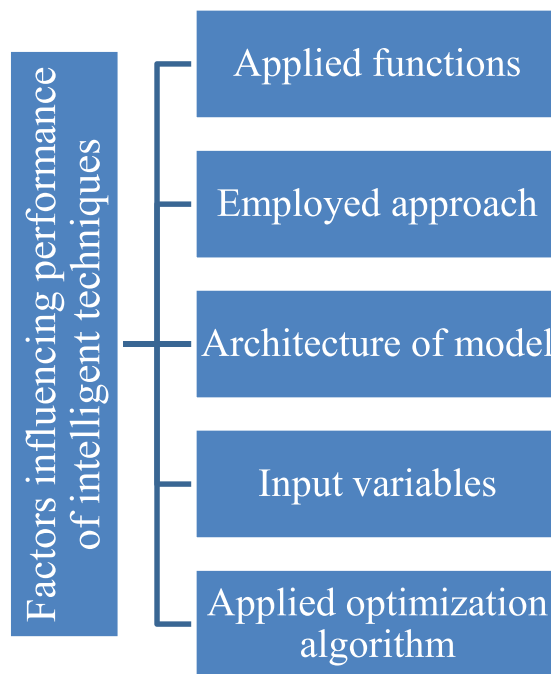


Fig. 1. Some of the most important factors influencing the performance of intelligent techniques.

local momentum, pressure gradient, surface information and turbulent viscosity were applied as inputs.

Regarding the ability of machine learning approaches in forecasting the complicated phenomena, boiling heat transfer characteristics, including heat transfer coefficient, Nusselt number etc., could be properly predicted and modeled by these techniques, which have been applied by different scholars in recent years. The present work reviews studies on the modeling of the boiling heat transfer by using machine learning methods. Since numerous constituents are involved in the performance and exactness of these models, studies on features of the models such as architecture and applied functions are considered here to provide a comprehensive and applicable reference for the scholars working on this subject. To reach this goal, studies on the relevant topics are gathered; subsequently, their findings are reviewed and discussed. In the final stage, some recommendations are proposed for the upcoming researches focusing on the similar subjects.

2. Applications of machine learning in modeling boiling characteristics

As denoted previously, intelligent techniques are usable for modeling different thermal systems. The performance, including exactness and comprehensiveness, is dependent on some factors. The main important ones are represented in Fig. 1. In the researches on the boiling modeling by these techniques, different aspects have been considered. Some of them have focused on the development of the most accurate model by testing different approaches and functions while some others have concentrated on proposing comprehensive model with applicability for different cases.

2.1. Boiling of conventional fluids

Boiling process is a complicated phenomenon and it is highly affected by different elements. In Fig. 2, some of the most important factors influencing this phenomenon are depicted. Regarding its complexities, it is crucial to apply powerful tools such as intelligent techniques for the modeling.

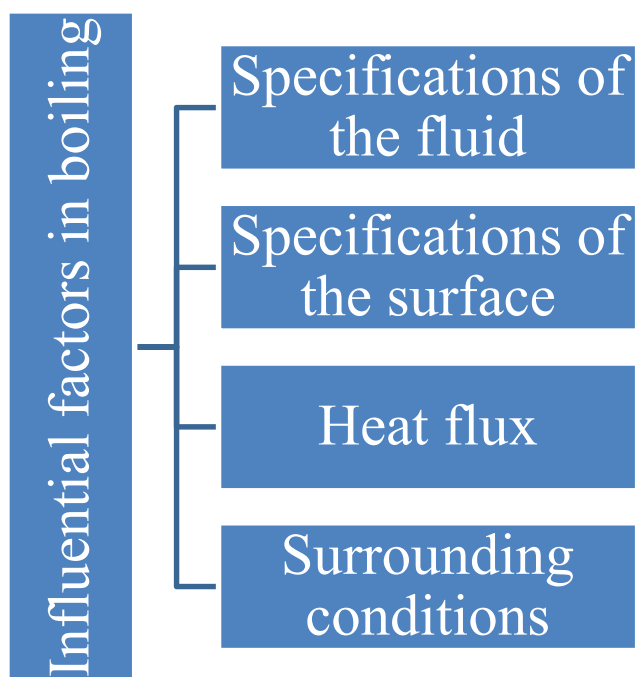


Fig. 2. Some of the important factors influencing boiling.

Different characteristics of the boiling phenomenon such as the critical heat flux, Heat Transfer Coefficient (HTC) and heat transfer enhancement can be predicted and modeled by using machine learning methods [36]. For instance, Liu et al. [37] employed an intelligent technique based on NN to model boiling heat transfer enhancement in the case of using additives. In their designed network, molecular structures of the additives were defined as the inputs. They found that the proposed model was able to predict heat transfer enhancement of boiling in the majority of cases with additives. In other research, Bouali et al. [38] applied ANN to forecast the HTC of inclined channel flow boiling. Quality and heat flow in addition to mass flux and orientation were utilized as the inputs in their model. Their model was very accurate for this purpose with a R^2 of 0.9993. In a work done by Wen et al. [39] a Radial Basis Function (RBF) NN was applied to model boiling heat transfer inside a horizontal tube with a smooth surface. The working fluid that was considered in their study was a refrigerant, R407C. The Root Mean Squared Error (RMSE) of the model for the case study was 10.9%, indicating acceptable outputs of the applied technique. In another work [40], an ANN was employed to forecast the minimum film boiling temperature for different rods quenched in water. In this case, they utilized subcooling temperature, thermophysical properties of substrate, ratio of length to diameter and saturation pressure as the inputs. The Mean Absolute Error (MAE) of the model was around 5%, which reveals its significant performance for this purpose.

In addition to exactness of the model, its comprehensiveness has high importance. It would be beneficial to develop more comprehensive models with wider applicability in order to forecast boiling in different conditions with various fluids. The models could be enhanced in terms of their comprehensiveness by considering more inputs. Within this context, using dimensionless numbers would be useful in order to apply different variables as the inputs. For instance, Wang et al. [41] applied four kinds of dimensionless parameters that were used in the previous studies as the inputs. Their models considered R22 and its alternatives refrigerant such as R410A, R407C and R134a. They found that the inputs of the models affect the precision of prediction and in the best case the root mean squared deviations of the model was 20.3%. In another work, Sajjad et al. [42] proposed a model by considering different properties of the surface including surface material (thermal

conductivities in range of 1.14 - 385 W/m.K), roughness (0.02–10.3 μm) and inclination of the surface (0–180°) in addition to saturation temperature (–196–100 °C) and pressure (25–400 kPa). Furthermore, thermophysical properties of the liquid and heat flux were used as the inputs. They focused on the HTC of nucleate boiling. In their study, different architectures of the network were applied and in the optimum condition, a R^2 of 0.994 is reached with a MAE of 0.65, revealing high performance of the proposed model. Compared with other previously proposed models, the predicted values by the DNN were much closer to the experimental data as shown in Fig. 3. In another work, Qiu et al. [43] proposed an ANN-based model for saturated flow boiling HTC of mini/micro channel on the basis of universal consolidated data. Several dimensionless numbers such as Bo, Bd, Co etc. were applied as the inputs of their network. The MAE of their model under optimal conditions was 14.3%.

Aside from the models proposed for forecasting the characteristics of boiling in conventional conditions without any additional limits or components affecting the boiling, some efforts have been undertaken to develop them for complex cases. In addition to conventional problems with boiling phenomenon, intelligent methods are applicable for modeling boiling in complicated conditions [44]. For instance, Cong et al. [45] applied an ANN to forecast the critical heat flux in saturated forced convective boiling on a heated surface with impinging jet as shown in Fig. 4. The inputs of the model were the ratio of liquid density to vapor density, reciprocal of the Weber number, L/D and impinging jet number. The RMSE of their model was 17.39%. In another study [46], different NNs and an Adaptive Neuro-Fuzzy Inference System (ANFIS) were used to model flow boiling HTC over tube bundles. The inputs of the model were vapor quality, mass flux, pressure and the bundle pitch. It was observed that using general regression NN (GRNN) provides the highest accuracy, followed by Radial Basis Network (RBN) and ANFIS, respectively. In another work [44], the ANFIS method with various kinds and numbers of membership functions were applied to model HTC in flow boiling over bundle of tube. The optimal model was able to predict the values within deviation of $\pm 5\%$. Calati et al. [47] used an ANN to model the Nusselt number in case of water pool boiling in metal foam. For this purpose, five inputs including pore density parameter, Prandtl number on the basis of effective thermal conductivity, porosity ratio, surface tension to buoyancy ratio and enthalpy ratio were considered. The forecasted data in the case of employing the model for the Nusselt number were within 10% of the actual values obtained in experimental work with a Mean Average Percentage Error (MAPE) of 9.55% while applying the proposed models in the previous study [48] led to a MAPE of 114%.

In the majority of the cases, boiling is considered for investigation for the fluids with medium boiling temperatures since they are usable for more applications; however, boiling characteristics of the fluids with low boiling point has its own importance since they can be applied for thermal management in very low operating conditions. Cryogenic heat pipes are among the devices applied for thermal management at very low temperatures [49]. Besides the refrigerants and other conventional fluids such as water with low or medium boiling temperatures, intelligent approaches are usable for modeling the boiling of fluids with very low temperatures such as hydrogen. For instance, Kuang et al. [50] used an ANN to model the nucleate flow boiling HTC of saturated hydrogen. They applied different combinations of dimensionless numbers to find the best case, as shown in Table 1, and found that using a combination that includes Bo, Ja, Pr and Re_{fo} (liquid only Reynolds number) are sufficient for capturing the changes of heat transfer capacity in the nucleate boiling of saturated hydrogen. It is noted that the Nusselt number of two-phase flow is derived from the wall superheat, and as such considering the Jacob number would lead to the superheat degree appearing on both sides of the determination equation, leading to the former being an unnecessary inclusion. Consequently, they further decreased the number of variables. In this case, the R^2 of their proposed model was 0.973 and MAE was 8.24% as shown in Fig. 5.

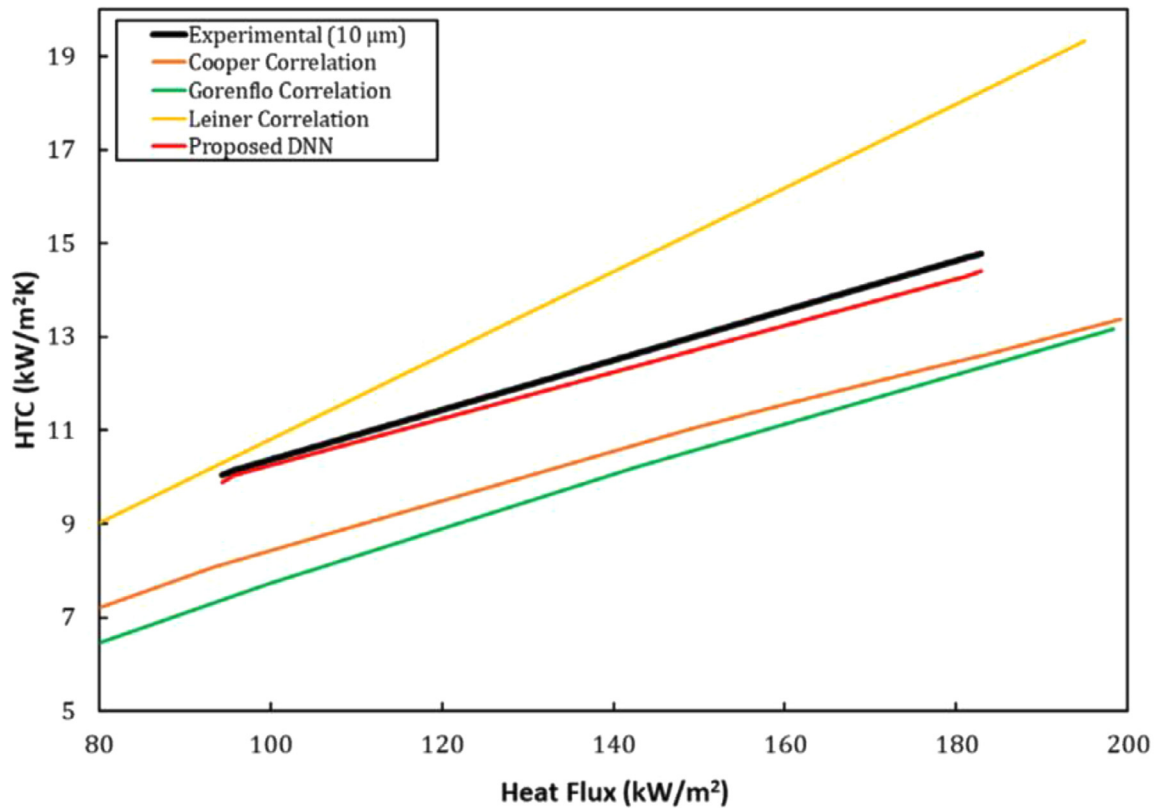


Fig. 3. Comparison of different models for predicting boiling heat transfer coefficient in surface roughness of 10 μm and working fluid of FC-77 [42].

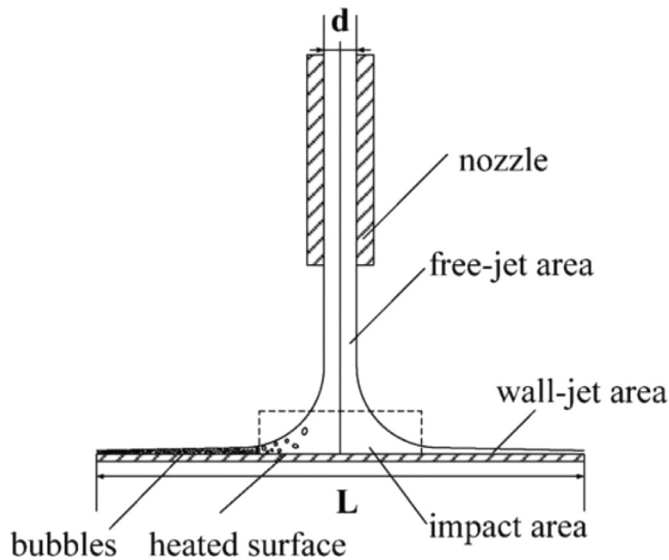


Fig. 4. Architecture of impinging jet [45].

In addition to the abovementioned approaches, as some of the most common kinds of intelligent techniques used for modeling of thermal system, other methods such as the ones based on fuzzy logic can be applied for exact prediction of boiling heat transfer characteristics [51]. For instance, Mansour et al. [52] applied fuzzy logic to determine the HTC of pool boiling based on experimental data; they focused on the boiling of R-134a at different conditions. They considered surface roughness, heat flux and working pressure as the inputs of the model. According to the comparison between the model outputs and experimental data, it was observed that the model was more accurate at low heat flux

Table 1

ANN results for various input dimensionless numbers [50].

Input dimensionless numbers	MAP (%)	R ²	R _{adj} ²
Ja, Bo, x, d _r , v _r , Pr, We, X _{tt} , Re _l , Re _{fo} , Pr, Bd, v _{r,w} , l _d	7.06	0.986	0.985
Ja, Bo, x, d _r , v _r , Pr, We, X _{tt} , Re _l , Re _{fo} , Pr, Bd, v _{r,w}	6.38	0.986	0.985
Ja, Bo, x, d _r , v _r , Pr, We, X _{tt} , Re _l , Re _{fo} , Pr, Bd	7.10	0.983	0.983
Ja, Bo, x, d _r , v _r , Pr, We, X _{tt} , Re _l , Re _{fo} , Pr	7.07	0.987	0.986
Ja, Bo, x, d _r , v _r , Pr, X _{tt} , Re _l , Re _{fo} , Pr	7.96	0.983	0.983
Ja, Bo, x, d _r , v _r , X _{tt} , Re _l , Re _{fo} , Pr	6.56	0.986	0.986
Ja, Bo, x, v _r , X _{tt} , Re _l , Re _{fo} , Pr	5.19	0.991	0.991
Ja, Bo, x, v _r , Re _l , Re _{fo} , Pr	4.54	0.994	0.994
Ja, Bo, x, v _r , Re _l , Pr	7.35	0.983	0.983
Ja, Bo, x, Re _l , Pr	8.30	0.983	0.983
Ja, Bo, Re _l , Pr	7.70	0.981	0.981
Ja, Bo, Re _{fo} , Pr	5.00	0.992	0.992
Bo, Re _{fo} , Pr	8.24	0.973	0.973
x, Re _{fo} , Pr	27.9	0.620	0.617
Ja, Re _{fo} , Pr	23.0	0.807	0.805
X _{tt} , Re _{fo} , Pr	35.9	0.622	0.619
Re _{fo} , Pr	50.4	0.427	0.424

x: vapor quality/input, X_{tt}: Martinelli factor, v_r: viscos50.4ity ratio, d_r: liquid to vapor density ratio.

with a deviation margin of $\pm 2\%$ while this value increases to $\pm 6\%$ at high heat fluxes. Higher deviation for increased heat fluxes was attributed to the impact of heat loss in this condition which could not be taken into account by the proposed model.

Exactness of the models on the basis of intelligent techniques is dependent on several factors; however, model architecture is one of the most significant ones. In ANNs, number of neurons and layers is very crucial in selection of the most accurate model. It is indicated that a network with three layers including one hidden layer in addition to the input and output layers is a universal approximator and is able to properly handle the majority of problems [53]; however, more com-

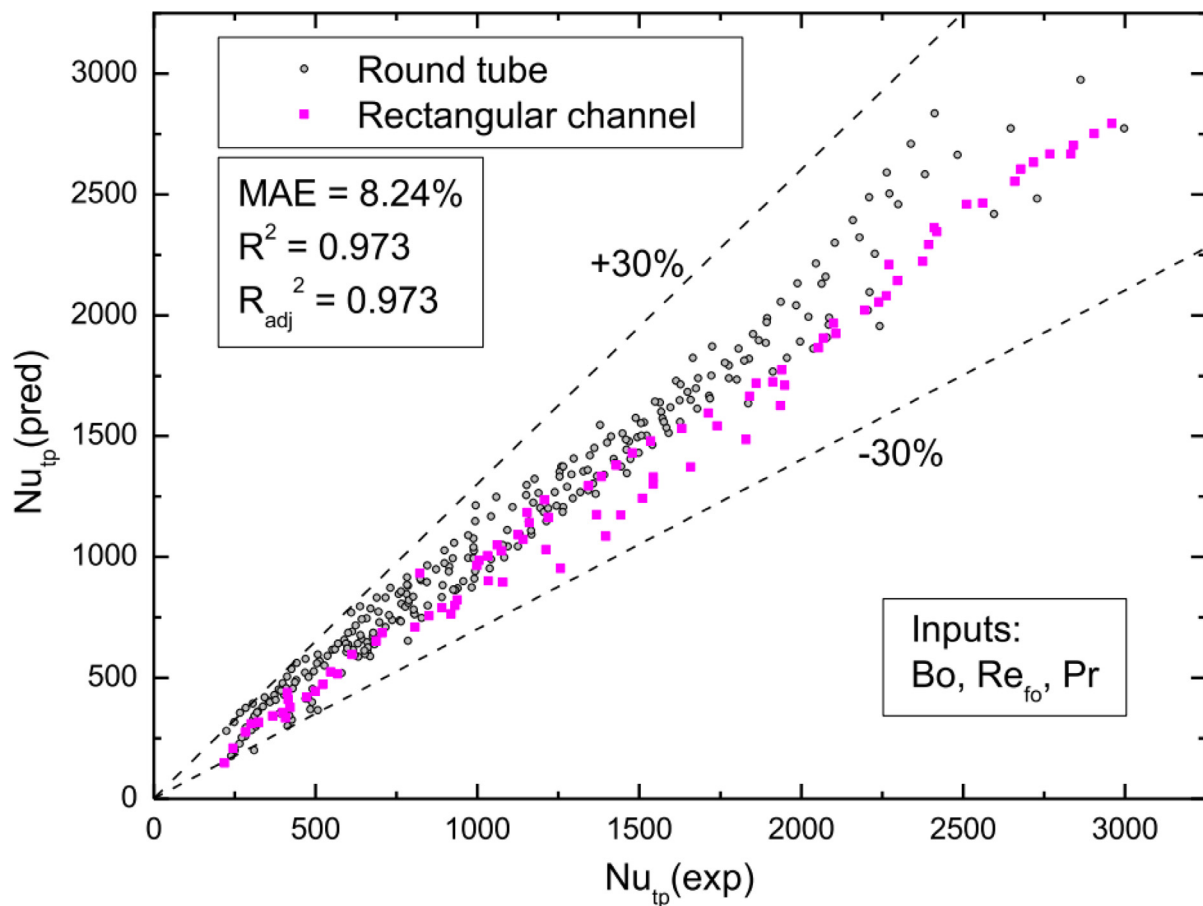


Fig. 5. Experimental and predicted values of Nusselt number of hydrogen flow boiling [50].

plex systems may require a higher number of hidden layers. Since the physics of boiling is complex in some cases, utilizing more than one hidden layer can be useful to achieve higher accuracy. In term of the number of neurons there is a tradeoff: increasing the number of neurons can cause overfitting while a reduction in the number of neurons can negatively affect the generalization capabilities of the model [54]. In fact there are many studies that have investigated the impact of a network's structure on their performance within the context of applying intelligent techniques to the modeling of boiling. As an example, Sajjad et al. [55] applied different numbers of neurons for a model proposed for predicting boiling HTC of porous surfaces. In their study, deep NN with three hidden layers and various numbers of neurons were tested and it was observed that using 30 neurons provided the highest value of R^2 . Similar to ANN-based models, the performance of other approaches is dependent on the architecture of the model and the methods used in the structuring of the algorithms. For instance, Das et al. [56] found that for predicting the HTC of water boiling through a zero-order adaptive fuzzy model, the increment in the number of membership functions would improve the precision. In another work, which was carried out by Das et al. [57] on the modeling of organic liquid pool boiling, it was found that in case of using a zero order fuzzy model and applying 6 member functions one obtains higher exactness in comparison with utilizing only 2 functions. In a study done by Mehrizadeh et al. [58], boiling HTC of various refrigerants in different types of tubes including low fin, plain tube, Thermoexcel-E and Turbo-B was modeled by utilizing different approaches such as ANN, Extreme Learning Machine (ELM), SVM and ANFIS. They found that deviation range was the lowest in case of employing ANN that was followed by ANFIS, SVM and ELM. Aside from the exactness, performance of the models were compared according to the training time and it was noticed that training time of

the ANN was the minimum one among the applied techniques. Based on both exactness and speed of training, ANN was selected as the most appropriate techniques.

2.2. Boiling of nanofluids

Nanofluids, consisted of the solid materials in nanometer dimensions and a base fluid [59,60], are promising fluids for heat transfer augmentation and efficiency improvement [61,62]. Nanofluids can be categorized by using different criteria; however, considering the type of nanostructures for categorization is very conventional. In this regard, nanofluids can be divided into two main groups: the ones containing single nanomaterial and the ones containing nanostructures composed of two or more materials, known as hybrid nanofluids. The properties of the nanofluids involved in their heat transfer characteristics as provided in Figure .

Aside from conventional fluids, boiling characteristics of nanofluids has high importance due to their utilization in two-phase heat transfer mediums. Boiling characteristics of these fluids have been investigated in different works and various intelligent techniques have been used for the modeling [63–66]. As an example, Hassanpour et al. [67] performed a study on the modeling of pool boiling heat transfer of Al_2O_3 /water nanofluid. They considered the concentration of the particles and their size, wall superheat and pressure as the inputs. In their work, different networks including RBF, Cascade Forward (CF) and MLP were tested, and it was observed that in the optimal case, MLP, a R^2 of 0.9929 is reached and an average absolute relative deviation of 9.53 obtained. In another work, Peng et al. [68] modeled wall superheat and boiling heat coefficient of Fe_3O_4 /water nanofluid by concerning different inputs including concentration of the nanofluid, heat flux and rough-

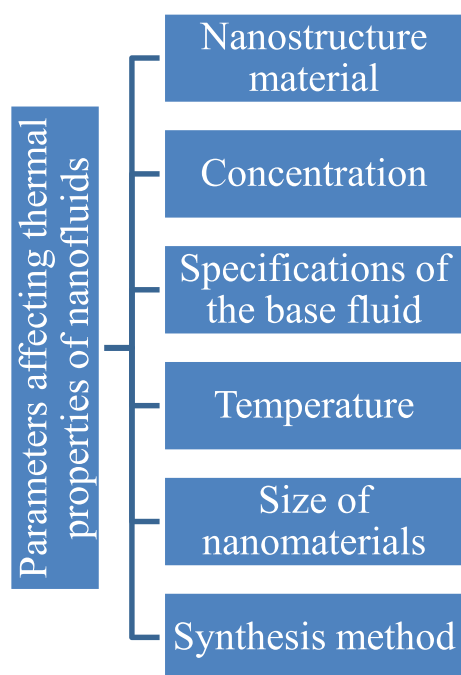


Fig. 6. Most important factors affecting thermal properties of nanofluids.

Table 2
Summaries of the reviewed works.

Reference	Fluid	Approach	Key Findings
Liang et al. [36]	R245fa	GRNN	Boiling HTC inside the horizontal tube was predicted with RMSE of 10.51%.
Liu et al. [37]	–	ANN	Boiling heat transfer improvement in cases of using additives was forecasted by the model.
Bouali et al. [38]	Water	ANN	HTC was predicted with R^2 of 0.9993.
Wen et al. [39]	R407C	RBF	RMSE of the model was 10.9%.
Bahman et al. [40]	Water	ANN	Minimum film boiling temperature was predicted with MAE of around 5%.
Wang et al. [41]	R22, R410A, R407C and R134a	ANN	The applied dimensionless numbers as the inputs influenced the precision of the model.
Sajjad et al. [42]	Refrigerants, water and dielectric liquids	ANN	For the optimum structure, R^2 of the model reached 0.994.
Qiu et al. [43]	Different fluids	ANN	MAE of the model was 14.3%.
Swain et al. [44]	Isobutane	ANFIS	Deviation range of the predictions was around $\pm 5\%$.
Cong et al. [45]	Water, R113 and R12	ANN	RMSE of the model was around 17%.
Swain et al. [46]	–	Different NNs and ANFIS	Using general regression NN provided the highest precision.
Swain et al. [44]	–	ANFIS	The optimal model was able to predict values within $\pm 5\%$ deviation.
Calati et al. [47]	Water	ANN	The forecasted Nusselt numbers were within 10% of experimental values.
Kuang et al. [50]	Hydrogen	ANN	R^2 of the model by considering Pr, Bo and Re_{fo} was 0.973.
Dadhich et al. [65]	Al_2O_3 /water and TiO_2 /water	ANN and correlation	ANN outperforms the correlation for predicting HTC.
Zendehboudi et al. [70]	–	ANN, MLR, LSSVM and ANFIS	ANN outperform in modeling nanoparticle migration ratio in the process of pool boiling.
Mansour et al. [52]	R-134a	Fuzzy logic	The model was able to forecast the HTC with higher accuracy at lower heat fluxes.
Sajjad et al. [55]	Different fluids	Deep NN	Neurons' number affect the performance of the model.
Das et al. [56]	Water	Adaptive neuro fuzzy	Increase in number of membership function would enhance the precision.
Das et al. [57]	Organic liquids	Different fuzzy approaches	Using 6 membership functions in the zero order fuzzy model led to higher accuracy compared with 2 membership function.
Mehralizadeh et al. [58]	Different refrigerants	ANN, ANFIS, SVM and Extreme Learning Machine (ELM)	Deviation range of the predictions was the lowest in case of using ANN.
Zendeboudi et al. [72]	Refrigerant oil mixtures with nanoparticles	BP-ANN, PSO-ANN, GA-LSSVM, CSA-LSSVM, GA-LSSVM, GMDH, Hybrid ANFIS, CHPSO-ANFIS and GA-RBF	GA-RBF outperformed other approaches.
Hassanpour et al. [67]	Al_2O_3 /water	ANN	R^2 of the model in optimum condition was 0.9929.
Peng et al. [68]	Fe_3O_4 /water	ANN	Both wall superheat and boiling HTC were accurately predicted.
Mehralizadeh et al. [69]	Water and Water-EG with TiO_2 - SiO_2 nanomaterials	ANN and ANFIS	Using ANN led to higher forecasting precision.
Saei et al. [73]	Refrigerant oil containing nanoparticles	ANFIS with different optimization algorithms	Using PSO-ANFIS provided the highest exactness.

ness. Their model was very accurate in predicting both outputs with a R value of 0.9986 and 0.99936 for wall superheat and boiling HTC, respectively. Besides conventional nanofluids, hybrid types, consisting of the nanostructures of two or more materials, have been tested for boiling and modeled. As an example, Mehralizadeh et al. [69] applied two intelligent methods including an ANN and Adaptive Neuro Fuzzy Inference (ANFIS) for modeling the boiling HTC of hybrid nanofluids with TiO_2 - SiO_2 nanomaterials. It should be mentioned that the base fluids were water and water-EG and the boiling surfaces were conventional and plain or modified by applying circular microchannels and circular microchannels with intersection lines. In their models, the concentration of nanofluid, heater surface type, ratio of water to water-EG mixture in the base fluid and ratio of TiO_2 to TiO_2 - SiO_2 were used as inputs. They observed that the forecasting accuracy of the ANN model was higher compared with the ANFIS. Other characteristics of nanofluids boiling could be modeled by employing intelligent methods. For instance, Zendeboudi et al. [70] applied different techniques including ANN, ANFIS, Least Squared Support Vector Machine (LSSVM) and Multiple Linear Regression (MLR) to predict migration ratio of nanoparticles in the process of nanorefrigerants' pool boiling. They found that an ANN outperforms other approaches with a R^2 of around 0.9994.

Aside from the architecture, functions and optimization methods are influential on the quality of the outputs. Optimization methods are mainly used in intelligent techniques to tune the hyperparameters and minimize the errors [71]. Regarding the difference in the performance of optimization approaches, the exactness of the models is influenced depending on the method used for regularization of the intelligent technique parameters. In a work [72], different techniques including Back

Propagation (BP) ANN, Particle Swarm Optimization (PSO) ANN, Genetic Algorithm Least Square SVM (GA-LSSVM), Coupled Simulated Annealing LSSVM (CSA-LSSVM), GA-LSSVM, Group Method of Data Handling (GMDH), Hybrid ANFIS, Conjugate Hybrid PSO-ANFIS (CHPSO-ANFIS) and GA- Radial Basis Function (GA-RBF) were tested for modeling nucleate pool boiling HTC of refrigerant oil mixtures with nanoparticles. They found that GA-RBF shows better performance in addition to higher integrity in the process of test and all data stages. The optimization algorithm coupled with the intelligent method is another factor influencing the performance of the method. In a study by Saeed et al. [73] different optimization algorithms including PSO, GA, Ant Colony Optimization (ACO), Differential Evolution (DE) were coupled with ANFIS to forecast pool boiling heat transfer. It was found that using PSO for optimization led to the model with highest precision with a R^2 of 0.998. Differences in the exactness of these models, proposed based on ANFIS, can be attributed to the different performances of the optimization approaches used for regulation of the parameters.

In Table 2, the findings of the reviewed studies are summarized.

3. Recommendations for upcoming studies

Although there are several valuable works in modeling the characteristics of boiling by employing intelligent methods, there are some points that could be considered in future works. First of all, the effect of model architecture on its precision and performance can be more extensively investigated in upcoming works in the relevant field [74]. Furthermore, more inputs could be defined in future work to provide more detailed comprehensiveness. For instance, features of the boiling surface and operating conditions are among the factors that could be more widely investigated in future works. It is also suggested that the coupling of various novel optimization techniques be considered to further enhance the precision. These optimization algorithms can be used to adjust the hyperparameters used in the intelligent methods.

Despite there being several works on the modeling of nanofluid boiling, hybrid nanofluids have not been considered widely. Hybrid nanofluids are composed of two or more nanomaterials that have been applied in various thermal systems and mediums such as solar and heat exchangers [75,76]. In some research, it has been shown that applying hybrid nanofluids could improve the heat transfer [77–79], making them appropriate as two-phase heat transfer mechanisms. Regarding some of the favorable features of hybrid nanofluids in terms of heat transfer, it would be useful to develop intelligent techniques for modeling the boiling of these fluids under various conditions.

4. Conclusion

The present article reviewed various works on the use of intelligent methods for the modeling of boiling characteristics. These techniques have shown significant performance in modeling different aspects of boiling; however, their performance could be affected by various elements. The main findings are summarized as follows:

- Intelligent techniques are applicable for modeling various characteristics of the boiling process including the heat transfer coefficient, wall superheat and minimum film boiling temperature.
- Model inputs could significantly influence the precision. A more extensive consideration of all influential factors would lead to increased exactness.
- By using dimensionless numbers, it would be possible to propose comprehensive models with applicability for different conditions.
- In addition to the inputs, the type of machine learning approach used for modeling would affect the precision of the outputs.
- Model architecture such as the number of membership function in ANFIS models or number of neurons in ANN affect the performance of the models proposed for boiling.
- Optimization algorithm coupled with intelligent methods influence the errors of the models.

- In addition to conventional fluids and refrigerants, intelligent techniques are able to model the boiling characteristics of nanofluids.

Declaration of Competing Interest

There is no conflict of interests between the authors.

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